

Alexander Laroche

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PhD Candidate, Department of Astronomy & Astrophysics

Research Interests

My research is focused on massive binary evolution, through a combination of observations and astrostatistics. I study observed post-interaction binaries, such as binary-stripped helium stars. I also develop data-driven models to discover new, rare stars in large-scale surveys. I have previously worked on dark matter substructure with strong gravitational lensing and 21 cm signal constraints during the Epoch of Reionization.

Education

University of Toronto

PhD in Astronomy & Astrophysics

Advisors: Maria Drout, Joshua S. Speagle

Title: Constraining binary evolution through discovery and characterization of post-interaction binaries in large-scale surveys

Toronto, ON, Canada

2022 – Present

McGill University

HBSc. in Physics (First Class Honours)

Advisor: Adrian Liu

Title: Quantifying density-ionization correlations with the 21cm power spectrum while including X-ray heating effects

Montreal, QC, Canada

2018 – 2022

Awards & Fellowships

NSERC Canada Graduate Scholarships - Doctoral (CGS-D)

120,000 CAD, Natural Sciences & Engineering Research Council

Sep 2024 – Aug 2027

Data Sciences Institute (DSI) Doctoral Student Fellowship

75,000 CAD, University of Toronto Data Sciences Institute

Sep 2024 – Aug 2027

NSERC Canada Graduate Scholarships - Master's (CGS-M)

17,500 CAD, Natural Sciences & Engineering Research Council

Sep 2023 – Aug 2024

Graduate Entrance Scholarship

5,000 CAD, David A. Dunlap Department of Astronomy & Astrophysics

Sep 2022 – Aug 2023

NSERC Undergraduate Summer Research Award (USRA)

9,600 CAD, Natural Sciences & Engineering Research Council

May 2021 – Aug 2021

Canada 150 Research Chair Undergraduate Research Grant

6,000 CAD, Canada 150 Research Chairs Program

May 2020 – Aug 2020

Charles River Laboratories Scholarship

2,000 CAD, McGill University

Sep 2020 – Aug 2021

Wing Hing Chan Scholarship in Science

500 CAD, McGill University

Sep 2020 – Aug 2021

Dean's Honours List

Top 10% of Faculty of Science, McGill University

Sep 2020 – Aug 2021

Publication List

Summary: 2 papers (+ 1 conference abstract) as first author, 3 as contributing author.

Peer-Reviewed Journal Publications.....

1. **Laroche, A.**, Speagle, J.S. "Closing the stellar labels gap: Stellar label independent evidence for $[\alpha/M]$ information in Gaia BP/RP spectra," **2025, ApJ, 979, 1.**
2. **Laroche, A.**, Gilman, D., Li, X., Bovy, J., Du, X. "Quantum fluctuations masquerade as halos: Bounds on ultra-light dark matter from quadruply-imaged quasars," **2022, MNRAS, 517, 2.**
3. Ludwig, B., Drout, M.R., Götzberg, Y., Land, D., **Laroche, A.** "The Stripped-star Ultraviolet Magellanic Cloud Survey (SUMS): The Ultraviolet Photometric Catalog and Stripped-star Candidate Selection," **2026, ApJ, 999, 1.**
4. Evans, F., Grondin, S., Ye, C., Webb, J.J., **Laroche, A.** & Bovy, J. "The ejection and detectability of high- and

hyper-velocity stars by compact object binaries in globular clusters,” 2025, in review at ApJ.

5. van Son, L.A.C., Yamaguchi, N., Nagarajan, P., Shenar, T., Sen, K., Laroche, A., Leiner, E.M., Sana, H., Pols, O.R., “Ongoing and Post-Mass-Transfer Binaries: A Living Catalog and Unified Review of Binary Mass Transfer Products” in review at ApJS.

Peer-Reviewed Conference Abstracts

1. Laroche, A., Speagle, J.S. “Closing the stellar labels gap: An unsupervised, generative model for Gaia BP/RP spectra,” 2023, ICML Workshop on Machine Learning for Astrophysics (6 pages).

Additional Research Positions

Undergraduate Research Assistant - UofT Galactic Astrophysics Group

2021-2022

Department of Astronomy & Astrophysics

University of Toronto

Supervisors: Jo Bovy, Daniel Gilman

Project: Constraining ultra-light dark matter with strong gravitational lensing

Undergraduate Research Assistant - McGill Radio Lab

2020-2021

Department of Physics

McGill University

Supervisors: Jonathan Sievers, Fernando Zago

Project: Data analysis for Probing Radio Intensity at high-Z from Mario (PRIZM)

Observing Programs

Successful Observing Proposals

[PI] Confirming a highly magnetic intermediate-mass helium star candidate

Gemini Observatory

(Awarded: 6 hours)

2026

[Co-I] Spectroscopy of WD-M dwarf binary candidates in star clusters

Lick Observatory

PI: Steffani Grondin (Awarded: 101 hours)

2025

[Co-I] Observational Constraints for Mass Transfer with SB2s

Las Campanas Observatory

PIs: Maria Drout & Ylva Götberg (Awarded: 16 hours)

2024

Observing Experience

6.5-meter Magellan Telescope [in-person and remote observing]

Las Campanas Observatory, Chile

PIs: Maria Drout & Ylva Götberg

2023, 2024

Spectroscopy of candidate binary-stripped helium stars

Selected Academic Presentations

Invited Talks/Seminars

Asteroseismology Group Meeting, KU Leuven

July 2025

Radial velocity survey of stars stripped in binaries in the Magellanic Clouds

Leuven, Belgium

Department Lunch Seminar, University of Toronto

March 2025

Radial velocity survey of stars stripped in binaries in the Magellanic Clouds

Toronto, Canada

Carnegie Tea Talk, Carnegie Observatories

December 2024

The first sample of orbital properties for stars stripped in binaries

Pasadena, California

Caltech Tea Talk, California Institute of Technology

December 2024

The first sample of orbital properties for stars stripped in binaries

Pasadena, California

Institute Seminar, Institute of Science and Technology Austria

July 2024

The first sample of orbital properties for stars stripped in binaries

Vienna, Austria

Contributed Talks

From discovery to a population: benchmarking stripped stars and companions

July 2024

Lightning talk - Non-conservative mass transfer in observed stripped star + MS binaries

Leiden, Netherlands

New Computational Methods in Milky Way Dynamics and Structure

July 2024

Talk - Modeling stellar spectra without stellar labels: a variational auto-encoder for Gaia XP spectra

Bavaria, Germany

Rare Gems in Big Data

May 2024

Talk - Discovering rare stellar populations in Gaia with stellar label independent modeling

Tucson, Arizona

Stable Mass Transfer in Binaries: from onset to remnants <i>Lightning talk - Non-conservative mass transfer in observed stripped star binaries</i>	March 2024 New York, USA
3,2,1: Massive Triples, Binaries and Mergers 2023 <i>Poster - An envelope lost in transport (Awarded best poster)</i>	July 2023 Leuven, Belgium
The Fortieth International Conference on Machine Learning <i>Poster - Closing the stellar labels gap</i>	July 2023 Honolulu, USA
Canadian Astro-Particle Physics Summer Student Talk Competition <i>Talk - Quantum fluctuations masquerade as halos</i>	Aug 2022 Subdury, ON
University of Toronto 2022 Stellar Stats Workshop <i>Talk - Constraining ultra-light dark matter by forward modeling flux ratios</i>	May 2022 Toronto, ON
Departmental Presentations, Seminars & Talks	
McGill University Undergraduate Research Project Presentation <i>Probing the quantum mechanics of ultra-light dark matter with strong lensing</i>	Dec 2021 Montreal, QC
University of Toronto NSERC USRA Poster Seminar <i>Probing the quantum mechanics of ultra-light dark matter with strong lensing</i>	Aug 2021 Toronto, ON
McGill University Undergraduate Research Thesis Presentation <i>Density-ionization correlations with the 21cm power spectrum including X-ray heating</i>	Apr 2021 Montreal, QC

Mentorship

Supervisor, Alejandro Ortega-Cruz Pietro <i>Project: Multi-modal learning for a fundamental understanding of variable stars</i> (Co-supervisors: Josh Speagle, Maria Drout)	May 2024 – Present
Undergraduate Mentor, University of Toronto <i>Advising undergraduate students on research and graduate school applications</i>	2022 – 2023

Teaching Experience

Introduction to Astrophysics <i>Graduate TA, University of Toronto</i>	Winter 2026
AST221: Stars and Planets <i>Graduate TA, University of Toronto</i>	Fall 2025
AST325/326: Introduction to/Practical Astronomy <i>Graduate TA, University of Toronto</i>	Fall 2024
AST222: Galaxies & Cosmology <i>Graduate TA, University of Toronto</i>	Winter 2024
AST199: Astronomy at the Frontier <i>Graduate TA, University of Toronto</i>	Fall 2023
AST201: Stars & Galaxies <i>Graduate TA, University of Toronto</i>	Winter 2023, 2025
AST101: The Sun and Its Neighbours <i>Graduate TA, University of Toronto</i>	Fall 2022
PAPER Tutor <i>Undergraduate math and physics tutor, Montreal, QC</i>	2019-2020

Selected Service & Outreach

Treasurer - UofT Graduate Astronomy Student Association <i>Manage finances for the Graduate Astronomy Student Association (GASA)</i>	2023 – Present
UofT Graduate Student Union Representative <i>Represent the interests of the astronomy grad student body within the Graduate Student Union (GSU)</i>	2022 – 2023
UofT AstroTours - Filmographer <i>Monthly organization of AstroTours, manage video content for organization</i>	2022 – 2023

Panelist – “Applying to grad school”

Fall 2022

UofT Astronomy Union

Reviewer for MNRAS, ApJ

2022 - Present